The unreasonable effectiveness of mathematics, solved: mathematics as a substrate-physical bootstrap from a single root

Eliot Gable*
Independent Researcher
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Wigner's 1960 observation that mathematics is "unreasonably effective" in describing nature rests on a tacit assumption: that mathematics is invented in pure abstraction and reality is a separate object onto which it is later mapped, with the success of the mapping requiring explanation. We propose that the puzzle dissolves when the assumption is dropped. In a substrate-physical framework in which a single ontological root — *the substrate exists* — is taken as the foundational hypothesis, every primitive of *the mathematics that physics uses* admits a substrate-physical reading: counting from event-counting, the natural numbers from extension-by-coupling-event, induction as a structural consequence of the substrate's iteration primitives, addition from linear superposition of substrate amplitudes in the free-propagation regime, multiplication capturing the substrate's non-linear cross-term coupling in its scalar commutative special case, division as the formal inverse with binding-energy substrate tension as the unmodelled remainder, the integers from opposing-tally processes, the rationals from observed ratios of counted quantities, the reals as Cauchy-completion against measurement, sets from substrate-type formation, topologies from measurement-precision neighbourhoods, smooth manifolds from elastic local flatness, differential forms from oriented coupling events, Lie groups from continuous symmetries, Lie algebras from infinitesimal symmetry generators, root systems from quantised standing-wave charges, and Coxeter groups from the substrate's reflection structure on its mode lattice. Each construction is paired with an observable discharge whose existence in laboratory or astrophysical data licenses the construction. On this reading, mathematics is not a separate language that happens to fit physics; it is the formal description of the substrate's primitive content, and its effectiveness in physics is not a coincidence to be explained but a structural consequence of the identity between description and described. We present the bootstrap chain end-to-end, identify the empirical anchor that the framework's E_8 root system shares with the observed Zamolodchikov spectrum in $CoNb_2O_6$, and discuss the open work of fully discharging the chain.

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I. INTRODUCTION

In his 1960 essay [1], Eugene Wigner asked why a body of abstract mathematical structure — complex numbers, Hilbert spaces, group representations, differential geometry — developed without reference to physics should turn

* 10015996+egable@users.noreply.github.com

out to describe physics so accurately that "it is difficult to avoid the impression that a miracle is involved." Two of his examples are paradigmatic. Heisenberg's matrix mechanics succeeded in spite of the fact that "the rules of matrix mechanics had been derived from problems which involved the old theory of the hydrogen atom"; complex numbers, used at the time mostly as a calculational convenience for real wavefunctions, turned out to be irreducible to quantum mechanics. The puzzle Wigner left us is sharp: how does a body of structure invented in pure abstraction fit a body of phenomena measured in experiments, when the two were not constructed in concert?

Wigner's question is usually treated in one of three ways. The *platonist* reply [2] grants mathematics independent reality and treats its match with physics as an observation about that reality. The *naturalist* reply [3, 4] treats mathematics as a refined extract from physical practice and argues the match is what one would expect. The *deflationary* reply [5, 6] questions whether the success is unreasonable at all, observing that the parts of mathematics most useful to physics tend to be the parts physics asked for. Each of these positions has its merits and its costs. None of them identifies a single technical commitment whose adoption would make the success of mathematics in physics a strict theorem rather than an unexplained tendency.

This paper proposes such a commitment. In a substrate-physical framework in which the foundational hypothesis is that *the substrate exists* — that there is a single ontological commitment to a coupled, event-counting, type-supporting substrate from which all observed phenomena unfold — we show that each primitive of the mathematics that physics uses is derivable from the substrate's primitive content, discharged by an observable phenomenon, and identical to the corresponding standard-mathematics object up to a bridge theorem (Sec. II A restricts the scope and lists what is *not* being claimed). The chain runs from the natural numbers through the rationals, the reals, sets, topologies, smooth manifolds, tensor calculus, differential forms, Lie groups, Lie algebras, root systems, and Coxeter groups; each is anchored in a specific substrate-physical process, and each carries an empirical signature that has already been observed.

On this reading, the answer to Wigner's question is structural and short. Mathematics is not a separate body of structure that happens to fit physics. The mathematics that appears in physics is, by construction, the formal description of the substrate's primitive content, and the success of that description on physical phenomena is a structural identity rather than a discovered fit. The "unreasonable effectiveness" of mathematics in the natural sciences ceases to be unreasonable when the mathematics-of-physics and the natural sciences are recognised as descriptions of a common source *at different levels of formalisation*. We do not extend this claim to mathematics that has no observed physical instantiation (see Sec. II A).

The paper is organised as follows. Section II introduces the framework's foundational ontology, identifying the single root commitment from which the bootstrap proceeds. Section III traces the chain from existence to the natural numbers and arithmetic, then shows that the four arithmetic operations have a parallel substrate-amplitude reading at the wave-pattern level. Section IV extends the chain to the integers, rationals, reals, sets, and topologies. Section V extends the chain through smooth manifolds, tensors, and differential forms. Section VI continues into Lie groups, Lie algebras, root systems, and Coxeter groups, and identifies the empirical anchor in $CoNb_2O_6$ that distinguishes the substrate's specific symmetry algebra from a generic one. Section VII states the resolution of Wigner's puzzle as a structural identity rather than a coincidence. Section VIII flags the limits of the present construction — which steps are observationally discharged, which are provisional, and what remains to be done.

II. THE FRAMEWORK'S FOUNDATIONAL ONTOLOGY

A. What this paper does and does not claim

Before stating the substrate's primitive content, we fix the scope of the argument, because several adjacent claims that the paper does *not* make would, if attributed to it, defeat it immediately.

a. What "substrate" means here. We use *substrate* as the framework's primitive ontological term for the single coupled, event-counting medium that everything observed *is* — not a backdrop on which phenomena live, but the thing of which all phenomena are configurations. The substrate is not given a formal mathematical definition in this paper. Such a definition would be circular within a paper deriving mathematics from substrate primitives; the substrate is the ontological commitment whose primitive content the paper's bootstrap describes formally. The reader who insists on a mathematical characterisation may treat "substrate" as a placeholder for "the empirical object whose properties are listed below" and read the paper as a structural-equivalence claim between that list and a specific bootstrap of standard mathematics.

b. What "single root" means here. The framework's implication graph terminates, transitively, in a single ontological commitment: that the substrate exists. It does not terminate in a single *rule*. The substrate's primitive content (Sec. II B) is rich — it includes classical inference, type formation, And/Or/Not-coupling, Test, Loop, extensionality, and infinity. Calling this single-rooted is a statement about the implication graph, not a claim that the substrate's content is a single primitive operation. The paper does not claim mathematics is bootstrapped from *nothing*; it is bootstrapped from the substrate's primitive content, and that content is what

makes the bootstrap go.

c. The hermeneutic circle. The bootstrap below uses higher-order logic, predicates, and quantification to describe substrate properties from which higher-order logic, predicates, and quantification are then said to fall out. This is the standard circularity of foundational projects (cf. Russell– Whitehead's use of logic to formalise logic, Lawvere's use of category theory to axiomatise category theory, the type-theoretic bootstrap of HoTT). It is not vicious, but it is real, and we note it explicitly: the paper's claim is that the substrate's primitive content is *coextensive* with what we call the metalogic, not that the metalogic is derived from a smaller starting point.

d. Distinction from the Mathematical Universe Hypothesis. The claim here is not Tegmark's Mathematical Universe Hypothesis [6]. The MUH says the universe *is* a mathematical structure, with all mathematical structures equally real. This paper's claim is narrower and asymmetric: the substrate has empirical content (specific particle masses, a specific symmetry algebra, specific topology) that selects one configuration out of the broader space of mathematical possibilities, and mathematics is what the substrate's primitive content looks like when described formally. The substrate is a particular thing; mathematics is its formal description; not every mathematical structure is a substrate, and the substrate's specific identity is empirical.

e. Mathematics beyond physics. We do not claim every mathematical object has a substrate-physical referent. Large cardinals, exotic forcing extensions, certain Banach–Tarski-type constructions, and other consequences of non-constructive choice principles have no observed substrate-physical instantiation. The paper's positive claim is restricted: each primitive of *the mathematics that physics uses* — elementary number theory, real analysis, point-set topology, smooth manifold theory, tensor calculus, exterior calculus, Lie theory, and the Coxeter classification — has a substrate-physical derivation and an observational discharge. Mathematics beyond that scope is neither denied nor explained here; it is consistent with the present construction that broader mathematics is generated by the closure of the substrate-derivable primitives under the framework's metalogic (extending well past anything observable), but we do not argue that here.

f. Bridge theorems. Each substrate-derived object below has both an existence-form definition (anchored in observation) and a constructive form (mirroring standard mathematics). The two presentations are connected by bridge theorems showing they describe the same object. The bridge theorems are stated and machine-verified throughout the framework's formal companion [11]; the present paper takes them as given and treats their full reproduction as out of scope. This is a load-bearing dependency, and we flag it as such.

g. Gödel-incompleteness. The substrate's primitive content (Sec. II B) suffices to interpret first-order Peano arithmetic, so any formal theory built from it inherits Gödel's first-incompleteness theorem: there exist arithmetical statements true in the standard model but not provable from the substrate-bootstrapped axioms. We do not treat this as a defect of the framework or as evidence against it. The incompleteness phenomenon is a property of any formal system rich enough to encode arithmetic, and the framework's substrate-physical discharge programme operates one level below the formal theory — it discharges *primitive content*, not the closure of all provable statements over that content. The Gödel sentence's substrate-physical status (whether it has any observable instance or is a purely logical artefact of the formalisation) is left as open metalogical work.

B. The substrate's primitive content

The framework's minimal ontology is committed to a single ontological root: the *substrate* exists. The substrate is a coupling-event-counting medium with the following primitive content, each piece of which is either grounded in the substrate's coupling-event definition or discharged by direct observation. We list the primitives because they are what the bootstrap will consume.

1. *Inference structure.* The substrate supports the standard logical inference rules: reflexivity, transitivity, application and abstraction congruence, beta reduction, modus ponens, hypothetical reasoning, antisymmetric deduction, variable instantiation, and type instantiation. Each rule is grounded in the substrate's coupling-event definition (a coupling event takes a state, applies an operation, resolves to a new state subject to admission criteria) and discharged by direct observation (chained-calibration agreement, biconditional substrate phenomena, exclusion-rule firing patterns).

2. *Foundational structure.* The substrate supports extensionality (identity-by-effect), eta-equivalence, selection from possibilities, the primitive operations Test, Loop, and And/Or/Not-coupling, and infinity in the sense that no finite bound is observed on the depth of recursive coupling.

3. *Bootstrap structure.* The substrate's event-counting satisfies two specific properties: *injectivity*, that extending two distinct event sequences each by one coupling event preserves their distinctness, and *non-zero successor*, that any sequence with at least one resolved event is distinguishable from the empty sequence. The substrate also supports stable *constant naming* (specific configurations retain identity across time and reference) and *type formation* (collections of configurations with a shared defining property are referred to as a kind).

These commitments are not framework conveniences. Each is a property of the universe that specific observ-

ables directly require: detector clicks have distinguishable counts, propagated light is distinguishable from unpropagated light, physical constants are constants, and electrons reproducibly share a single charge magnitude and mass.

We make explicit the commitment that does the work: every formal hypothesis the framework adopts carries an `impliedfrom` annotation listing the parent hypotheses it descends from, and a `discharge` annotation specifying the observational obligation that confirms it. Following the `impliedfrom` annotations transitively, every substrate hypothesis the framework commits to traces back to the single root: *the substrate exists.* The implication graph is, by construction, complete; what remains is the discharge work, which is empirical and collectively in progress [11].

III. FROM EXISTENCE TO ARITHMETIC

A. The derivation chain from *Exist*

The bootstrap from the framework's single ontological root (*exist*) to arithmetic proceeds through a chain of unpackings, each one a formal statement paired with a substrate-physical discharge. We list the chain in order; the rest of this section then formalises each step.

a. Step 1 — Exist. The root commitment is the bare verb-claim that something exists:

$$\exists s.\ \mathrm{Exist}(s). \tag{1}$$

Substrate-physically: every measurement ever made is at minimum a witness that something exists rather than nothing.

b. Step 2 — Substrate. The first unpacking commits to a substantive entity carrying the existence-property:

$$\exists \Sigma.\ \mathrm{Exist}(\Sigma)\ \wedge\ \Sigma \text{ is a substrate}, \tag{2}$$

where "substrate" is the framework's theory-neutral term for the single coupled medium that everything observed *is*, with phenomena as configurations of the substrate rather than objects on top of it. *Substrate-physically:* everything that exists *is* the substrate; observations are not attached to a separate medium, they are the substrate observing itself. The empirical witness is the universal commensurability of measurements across instruments and locations: every multi-instrument cross-validation in the empirical record is a witness that the same single substrate is the thing all instruments are configurations of.

c. Step 3 — States and operations. For an event to occur, the substrate must have states on which the event acts and operations producing different states. With state type $\mathcal{S}$ and operation type $\mathcal{O} = \mathcal{A} \to \mathcal{A}$ (where $\mathcal{A}$ is the substrate-amplitude type):

$$\exists s, t \in \mathcal{S}\ \exists o \in \mathcal{O}.\ \ o(s) = t\ \wedge\ s \neq t. \tag{3}$$

Substrate-physically: every laboratory observation is an operation transforming an initial state into a strictly different final state.

d. Step 4 — Time as change, coupling as the change-maker. Step 3 commits the existence of a state-change: an initial state s, a final state t, and an operation o with $o(s) = t$ and $s \neq t$. A state-change is, by construction, the difference between *before* and *after*: there is a state when the change has not yet happened (the initial state) and a state when it has (the final state). The before/after distinction is the substrate's notion of *time*: without time there could be no before and no after, hence no distinction between s and t, hence no change at all. And without change, no observations can be made: an observation is itself a coupling event, and nothing distinguishable can be recorded in a substrate where nothing is occurring. A void in which nothing changes and nothing observes is indistinguishable from *non-existence* (Step 1's $\exists s.\,\mathrm{Exist}(s)$ is exactly the denial that non-existence obtains). Hence the root commitment that something exists, together with Step 3's commitment that the existing thing admits state-changes, jointly force the existence of time. Time is therefore not an extra postulate added on top of the state-change commitment — it *is* change occurring, and the existence of even one state-change in the substrate forces time into the substrate's content. Formally, a change is a triple

$$\big(s_{\text{before}},\ o,\ s_{\text{after}}\big) \qquad \text{with} \quad o(s_{\text{before}}) = s_{\text{after}}, \quad s_{\text{before}} \neq s_{\text{after}}, \tag{4}$$

and the temporal ordering on the substrate is generated by the "before" relation between such triples (closing transitively through chains of operations).

For Eq. (4) to instantiate, something must *occur* that takes s_{before} to s_{after} — the operation o must actually run on the initial state and resolve to the final one. We give this thing a name: a *coupling event* is one elementary occurrence of an operation resolving on a state. "Coupling" is a label rather than an extra commitment; the existence of coupling events follows immediately from Step 3, since any particular state-change-triple Eq. (4) is, by definition, a coupling event.

The substrate's intrinsic unit of time at a location l is then the duration of one elementary coupling event there:

$$\tau_c(l) := T(l), \tag{5}$$

where $T : \mathcal{L} \to \mathbb{R}_{>0}$ assigns a positive duration to each substrate location $l \in \mathcal{L}$. The cumulative count of resolved coupling events at l is the integer-valued elapsed time at l in units of $\tau_c(l)$, and the substrate-physical content of "how much time has passed at l" is exactly that count. *Substrate-physically:* time is not a separate manifold smuggled in; it is what the substrate's state-change commitment forces, with coupling events as its elementary increments. (The use of $\mathbb{R}_{>0}$ as the codomain of T here is for exposition; the substrate-reals as the value-type for T are themselves derived in Sec. IV C, after which this typing becomes formally well-posed.)

e. Step 5 — Event sequences and event counting. Substrate events admit being arranged into ordered sequences (an event has a before and an after, given by the time it resolved relative to other events), and each sequence admits being counted. For an event-type $\mathcal{T}$, write $\mathcal{T}^*$ for the collection of finite ordered sequences of $\mathcal{T}$-events (including the empty sequence). The substrate commits the existence of a counting function $K : \mathcal{T}^* \to \mathcal{N}$ together with a designated zero $Z \in \mathcal{N}$ and a successor $S : \mathcal{N} \to \mathcal{N}$ satisfying:

$$\exists e \in \mathcal{T}^*.\ K(e) = Z \quad \text{(empty sequence)}, \tag{6}$$

$$\forall x, y \in \mathcal{N}.\ S(x) = S(y) \Rightarrow x = y \quad \text{(injectivity)}, \tag{7}$$

$$\forall x \in \mathcal{N}.\ S(x) \neq Z \quad \text{(non-zero successor)}. \tag{8}$$

Substrate-physically: detector clicks, pulsar pulses, and atomic-clock cycles all extend their event-sequences one resolved coupling event at a time without ever returning the count to zero and without two distinct sequences collapsing to the same count.

f. Step 6 — Iteration primitives. The substrate supports four primitives that together license the closure of countable iteration. Each is presented as a necessity-chain (the reason it *must* be supported, given Steps 1–5), then the equation, then the substrate-physical reading.

Test. Why it must hold: Step 3 commits the existence of distinct states $s, t \in \mathcal{S}$ with $o(s) = t \neq s$. For $s \neq t$ to be a substantive claim rather than vacuous notation, the substrate must have some way of *telling them apart* — some procedure that returns one answer on s and a different answer on t. That procedure is exactly predicate evaluation: a function that takes a configuration and returns yes-or-no. If predicate evaluation could not happen, the substrate could not register that $s \neq t$, hence could not register a change, hence could not have state-changes at all, contradicting Step 3 (and through Step 3, Step 4 and Step 1's $\exists s.\,\mathrm{Exist}(s)$ via the Step-4 non-existence collapse). Therefore predicate evaluation is the function

$$\textsc{Test} : \mathrm{Pred} \times \mathcal{S} \to \{\bot, \top\}, \quad \textsc{Test}(P, s) = P(s). \tag{9}$$

Substrate-physically: every measurement device, by definition, implements Eq. (9) for some predicate P ("has this threshold been crossed?"); the empirical record of distinguishable detector outcomes is the discharge.

Loop. Why it must hold: Step 5 commits an unbounded event-counting function $K : \mathcal{T}^* \to \mathcal{N}$ with injective successor S and no upper bound. For K to take values beyond a single application of S, the substrate must be able to *re-apply* S to its own previous output, repeatedly. "Re-apply the same operation" is the meaning of iteration. If iteration could not happen, S could be applied at most once; the substrate's counts would be limited to $\{Z, S(Z)\}$ with no $S(S(Z))$, contradicting Step 5's commitment that arbitrarily long event-sequences exist (and through it, contradicting the empirical existence of sequences with more than two events — every observed pulsar pulse-train, every detector run with more than one click, every atomic clock cycle). Therefore iteration of an operation under a $\textsc{Test}$ termination condition exists, defined by the recursive equation

$$\textsc{Loop}(o, P, s_0) = \begin{cases} s_0 & \text{if } \textsc{Test}(P, s_0) = \top, \\ \textsc{Loop}(o, P, o(s_0)) & \text{otherwise.} \end{cases} \tag{10}$$

Substrate-physically: every counted process running "until condition P holds" is the substrate executing Eq. (10); if P never holds the iteration runs without bound, licensed by infinity below.

Extensionality. Why it must hold: suppose two operations $o_1, o_2 \in \mathcal{O}$ acted identically on every state ($\forall s.\, o_1(s) = o_2(s)$) but were nonetheless counted as distinct in the substrate. The difference between them would then be *undetectable* by any state-change observation — it would make no difference to anything the substrate does. The framework's discharge programme (Sec. II A) requires every substrate commitment to have an empirical witness; an undetectable difference has none, and a commitment with no possible discharge cannot be admitted into the substrate's primitive content. Therefore operations are identified by their effect:

$$\forall o_1, o_2 \in \mathcal{O}. \quad \big(\forall s \in \mathcal{S}.\ o_1(s) = o_2(s)\big) \ \Rightarrow\ o_1 = o_2. \tag{11}$$

Substrate-physically: the laboratory cannot distinguish operations that produce identical results on every input; what the substrate cannot distinguish, it does not separate.

Infinity. Why it must hold: suppose there were a finite upper bound $N \in \mathcal{N}$ on event-counts, so that no event-sequence has more than N events. Then $S^N(Z)$ would be the maximum count, and $S(S^N(Z))$ would either fail to exist (contradicting that $S : \mathcal{N} \to \mathcal{N}$ is total) or equal some earlier count $S^k(Z)$ with $k \leq N$ (contradicting event-counting injectivity, Axiom III.3, since $S^N(Z) \neq S^{k-1}(Z)$ would force $S(S^N(Z)) \neq S(S^{k-1}(Z)) = S^k(Z)$). Either branch is a direct contradiction with Step 5. The bounded alternative is also empirically excluded: every alleged finite ceiling on event-counts has been exceeded by observation (atomic-clock cycles, photon-counts in long-baseline measurements, primordial light-front expansion in cosmology). Therefore

$$\forall N \in \mathcal{N}.\ \exists e \in \mathcal{T}^*.\ K(e) \succ N, \tag{12}$$

where $\succ$ is the substrate's order on counts inherited from S (every count has a strictly-greater successor count). *Substrate-physically:* the substrate has no event-counting ceiling; counted processes accumulate without an observed upper bound.

Together Eqs. (9)–(12) are forced by Steps 1–5, and they license the closure of the iteration $\{Z, S(Z), S(S(Z)), \ldots\}$ to the totality $\mathbb{N}$ formalised in Definition III.5.

g. Step 7 — Naturals, addition, multiplication. Why $\mathbb{N}$ must exist as a totality. Step 5 commits an unbounded

counting function $K : \mathcal{T}^* \to \mathcal{N}$; Step 6 commits the closure of $\{Z, S(Z), S(S(Z)), \ldots\}$ under Loop plus infinity. The codomain of K must *contain* every count the substrate ever produces. If $\mathbb{N}$ failed to exist as a single object, then either there would be some count the substrate produces that has no value in $\mathcal{N}$ (contradicting totality of K) or $\mathcal{N}$ would be bounded (contradicting infinity, Eq. (12)). The unique object that contains exactly $\{S^k(Z) : k \text{ produced by Loop}\}$ and nothing else is the initial $(\mathbf{0}, S)$-algebra (Definition III.5); its existence is therefore forced by Steps 5 and 6.

Why addition must exist as a total operation on $\mathbb{N}$. Substrate event-sequences compose by concatenation: if sequence e_1 has count m and sequence e_2 has count n, then the concatenated sequence $e_1 \cdot e_2$ has *some* count, and that count must be a definite element of $\mathbb{N}$ (else event-counting injectivity fails, since two concatenated sequences with the same parts could give different totals). The substrate's iteration primitive (Loop, Eq. (10)) applied to S starting from m for n iterations is exactly what produces this count. Therefore addition is the total binary operation defined by

$$m + \mathbf{0} = m, \qquad m + S(n) = S(m + n), \tag{13}$$

which is just the iteration of S packaged as a function of n. The recursion is well-defined and total because $\mathbb{N}$ is the initial $(\mathbf{0}, S)$-algebra: the universal property of Definition III.5 guarantees the existence and uniqueness of the function $n \mapsto m + n$ for each fixed m.

Why multiplication must exist as a total operation on $\mathbb{N}$. Substrate event-sequences also compose by *repetition*: observing a counted process n times, with each repetition contributing a count of m, must have a definite total count (else processes that should aggregate would have indeterminate results, again breaking injectivity — a detector observed across n identical windows would have an undefined cumulative count, but detectors *do* have cumulative counts). Repetition is iterated addition: add m to a starting count of $\mathbf{0}$ exactly n times. The substrate's Loop applied to addition-by-m for n iterations is what produces this. Therefore

$$m \times \mathbf{0} = \mathbf{0}, \qquad m \times S(n) = (m \times n) + m, \tag{14}$$

again well-defined and total by the universal property of $\mathbb{N}$.

Substrate-physically: a detector at click-count m that registers n more clicks ends up at $m + n$ (Loop executes S exactly n times); a detector observed across n measurement windows each registering m clicks has cumulative count $m \times n$ (Loop executes addition-by-m exactly n times). Without either operation, two perfectly observable physical processes would have undefined results — but they have observed results, so both operations must exist.

h. Step 8 — Subtraction (partial), then integers. Why subtraction must exist (at least partially) on $\mathbb{N}$. Substrate processes are observed in which a count *decreases*: an electric-charge accumulator can register both positive and negative contributions and end with a smaller magnitude than it had part-way through; a particle counter can have particles both added and removed; a stockpile of resolved coupling events can have the un-resolved difference asked of it. "Reverse the addition of n events" must therefore be a substrate-meaningful operation when its result is a non-negative count (because the reversed count is itself an observed substrate quantity, hence in $\mathbb{N}$). The left-cancellation lemma (Lem. III.14) guarantees that the result, when defined, is unique. Hence subtraction exists on $\mathbb{N}$ as the partial operation

$$m - n := \begin{cases} \text{the unique } k \in \mathbb{N} \text{ with } n + k = m & \text{if } n \le m, \\ \text{undefined} & \text{otherwise.} \end{cases} \tag{15}$$

Why $\mathbb{Z}$ *must exist as a totality.* Eq. (15) is partial: "remove 5 from 3" has no answer in $\mathbb{N}$. But substrate processes that net-remove more than they net-add do exist and *do* have definite outcomes: a charge accumulator that adds 3 and removes 5 has a definite final reading of -2 units; an account balance with deposits 3 and withdrawals 5 reads -2. If subtraction stayed partial, these observed processes would have no substrate-representable result, contradicting Step 5's discharge that all observed counted processes have definite values. The substrate must therefore extend $\mathbb{N}$ to a structure in which subtraction is total. The minimal such extension — the unique extension that adds inverses and nothing more — is the Grothendieck quotient: define

$$(a, b) \sim_{\mathbb{Z}} (c, d) \iff a + d = c + b, \tag{16}$$

(equivalently, (a, b) and (c, d) represent the same difference $a - b$ via an equation in $\mathbb{N}$ that doesn't require negative numbers to exist), and set

$$\mathbb{Z} := (\mathbb{N} \times \mathbb{N}) / \sim_{\mathbb{Z}}, \qquad [a, b] - [c, d] := [a + d,\ b + c]. \tag{17}$$

Theorem IV.5 verifies that $\mathbb{Z}$ is a commutative ring with $\mathbb{N}$ embedded as the non-negative integers and subtraction now total.

Substrate-physically: the discharge witness is opposing-tally processes — an electric-charge accumulator with positive and negative contributions, an account balance with deposits and withdrawals, a particle counter with added and removed tallies. The final readings of these processes are observed to take negative values; if $\mathbb{Z}$ did not exist as a totality, those readings would have no substrate value to land in.

i. Step 9 — Division (partial), then rationals. Why division must exist (at least partially) on $\mathbb{Z}$. Substrate processes are observed in which a count is *shared equally* across a number of bins: m events distributed across q identical channels gives m/q events per channel, and when this share is itself an integer (i.e. q divides m), the per-channel count is a definite element of $\mathbb{Z}$. The cancellation property of $\mathbb{Z}$ guarantees uniqueness when defined.

Hence division exists on $\mathbb{Z}$ as the partial operation

$$p \div q := \begin{cases} \text{the unique } k \in \mathbb{Z} \text{ with } q \times k = p & \text{if } q \neq 0 \text{ and } q \mid p, \\ \text{undefined} & \text{otherwise.} \end{cases} \tag{18}$$

Why $\mathbb{Q}$ must exist as a totality. Eq. (18) is partial: "share 7 across 3 channels" has no integer answer. But the substrate observes count-over-count ratios with non-integer values: "2 out of every 5 decays go to channel A" fixes a ratio $2/5$ that is not in $\mathbb{Z}$; multiplet positions of stable modes pick out lattice fractions like $1/3, 2/3$ for charge quantum numbers. If division stayed partial on $\mathbb{Z}$, these observed ratios would have no substrate-representable value, again contradicting Step 5's discharge for definite-valued counted processes. The substrate must therefore extend $\mathbb{Z}$ to a structure in which division by any non-zero element is total. The minimal such extension — the unique extension that adds multiplicative inverses to all non-zero elements and nothing more — is the field of fractions: define

$$(p, q) \sim_{\mathbb{Q}} (r, s) \iff ps = rq, \tag{19}$$

(equivalently, (p, q) and (r, s) represent the same fraction via an equation in $\mathbb{Z}$ that doesn't require non-integer ratios to exist), and set

$$\mathbb{Q} := (\mathbb{Z} \times (\mathbb{Z} \setminus \{0\}))/\sim_{\mathbb{Q}}, \qquad \tfrac{p}{q} \div \tfrac{r}{s} := \tfrac{ps}{qr} \quad (r \neq 0). \tag{20}$$

Theorem IV.8 verifies that $\mathbb{Q}$ is a field of characteristic 0 with $\mathbb{Z}$ embedded and division now total on $\mathbb{Q}^* = \mathbb{Q} \setminus \{0\}$.

Substrate-physically: the discharge witness is count-over-count processes — the number of decays in channel A over the total number of decays in a sample, the count of atomic-clock cycles per second-marker, the lattice index of a stable mode within a multiplet. The observed values of these processes include non-integer ratios that have no representation in $\mathbb{Z}$; if $\mathbb{Q}$ did not exist as a totality, those values would have no substrate type to land in.

j. Step 10 — Limits, then reals. Why limits must exist. Substrate measurements admit *indefinite refinement*: an atomic transition frequency, a relative decay rate, a mass ratio can each be measured to a definite rational value at finite precision, then re-measured to a closer rational at higher precision, and the substrate's infinity primitive (Eq. (12)) places no upper bound on the refinement depth. Each refinement is in $\mathbb{Q}$, but the *value being refined toward* (the actual measured quantity) need not itself be rational — and observation has long produced quantities (the speed of light in any unit system, the fine-structure constant, atomic spectral ratios involving irrational geometric factors) that are not rational. If the limit of a refinement sequence had no substrate-representable value, the substrate would have measurable quantities with no type to land in, contradicting Step 5's discharge.

Why Cauchy completion is forced. The substrate's Loop generates each refinement sequence (each step adds one further measurement); the substrate's extensionality (Eq. (11)) forces two refinement sequences that converge to the same limit to be identified (they witness the same measured quantity, so they must be the same substrate-real — if they were distinct, the substrate would carry an undetectable difference). The set of refinement sequences whose differences shrink to zero is exactly the Cauchy sequences; the equivalence relation that identifies sequences with zero limiting difference is exactly Cauchy equivalence:

$$\mathcal{C}(\mathbb{Q}) = \big\{(a_n)_{n \in \mathbb{N}} : \forall \varepsilon \in \mathbb{Q}_{>0}\, \exists N\, \forall m, n \geq N.\, |a_m - a_n| < \varepsilon\big\}, \tag{21}$$

$$(a_n) \sim_{\mathbb{R}} (b_n) \iff \forall \varepsilon \in \mathbb{Q}_{>0}\, \exists N\, \forall n \geq N.\, |a_n - b_n| < \varepsilon. \tag{22}$$

The substrate's commitments thereby force the totality

$$\mathbb{R} := \mathcal{C}(\mathbb{Q})/\sim_{\mathbb{R}}, \qquad \lim_{n \to \infty} (a_n) := [(a_n)] \in \mathbb{R}. \tag{23}$$

Theorem IV.13 verifies that $\mathbb{R}$ is a complete ordered field. Without $\mathbb{R}$, refinable measurements would either truncate at a finite precision (contradicting infinity) or produce values with no substrate type (contradicting the empirical existence of such measurements).

Substrate-physically: every measured physical quantity that admits indefinite refinement is the substrate-real $[(a_n)]$ where (a_n) is the sequence of refined rational measurements; $\mathbb{R}$ is forced into the substrate's content by the combination of Loop (refinement is iteration), infinity (refinement is unbounded), and extensionality (sequences converging to the same limit are the same substrate-real).

The remainder of Sec. III and Sec. IV formalises Steps 5–10. The intermediate steps (Exist, Substrate, States and Operations, Coupling-events-as-time) are not formal mathematical content themselves; they are the ontological commitments that license the formal content. Each carries its own discharge in the framework's formal record [11].

B. Counting and the natural numbers

We formalise Step 5 above, then extend through Step 7. The substrate-physical reading: zero is the count of the empty event-sequence; the successor operation increments the count by one coupling event; the substrate's Loop construct iterates the successor without bound.

Definition III.1 (Substrate-zero)**.** **0** *is the count of the empty substrate event-sequence: the initial state of every counted substrate process before any coupling event has resolved.*

Definition III.2 (Successor)**.** *There is an operator S on event-counts that maps a count n to the count obtained by extending its event-sequence by one resolved coupling event.* Substrate-physically: *every detector click that goes from m clicks to $m+1$ clicks instantiates S; the*

operator's domain and codomain are jointly fixed by the closure construction in Definition III.5 below.

The substrate's *Bootstrap structure* primitives (Sec. II B) commit the substrate to:

Axiom III.3 (Event-counting injectivity)**.** *S is injective:*

$$\forall m, n. \quad S(m) = S(n) \;\Rightarrow\; m = n. \tag{24}$$

Axiom III.4 (Non-zero successor)**.** *$\mathbf{0}$ is not in the image of S:*

$$\forall n. \quad S(n) \neq \mathbf{0}. \tag{25}$$

The substrate's Loop primitive iterates S without bound; its *infinity* primitive guarantees the iteration is not truncated.

Definition III.5 (Substrate-naturals)**.** *$\mathbb{N}$ is characterised by the universal property: for any set X equipped with an element $\mathbf{0}_X \in X$ and a function $S_X : X \to X$, there is a unique function $f : \mathbb{N} \to X$ with $f(\mathbf{0}) = \mathbf{0}_X$ and $f(S(n)) = S_X(f(n))$ for all $n \in \mathbb{N}$. Equivalently, $\mathbb{N}$ is the initial $(\mathbf{0}, S)$-algebra in the metalogic's universe of sets. The successor S becomes a function $S : \mathbb{N} \to \mathbb{N}$ by this characterisation, and Axioms III.3–III.4 fix its action.* Substrate-physically: *every counted process at every location instantiates a $(\mathbf{0}_X, S_X)$ pair (initial state plus a one-coupling-event step), and the unique map f is the substrate's* Loop *executed on that process.*

Theorem III.6 (Induction principle)**.** *For every predicate P on $\mathbb{N}$,*

$$\big[\, P(\mathbf{0}) \;\wedge\; \forall k \in \mathbb{N}.\, P(k) \to P(S(k)) \,\big] \;\Longrightarrow\; \forall n \in \mathbb{N}.\, P(n). \tag{26}$$

Proof. Let $X := \{n \in \mathbb{N} : P(n)\}$, equipped with the inclusion $\iota : X \hookrightarrow \mathbb{N}$ as its S_X-action lifted from S (well-defined since $\forall k.\, k \in X \Rightarrow S(k) \in X$ by hypothesis). Then X is a $(\mathbf{0}, S)$-subalgebra of $\mathbb{N}$ containing $\mathbf{0}$. Apply the universal property of Definition III.5 to the pair $(X, S|_X)$: there is a unique $f : \mathbb{N} \to X$ with $f(\mathbf{0}) = \mathbf{0}$ and $f(S(n)) = S(f(n))$. The composition $\iota \circ f : \mathbb{N} \to \mathbb{N}$ also satisfies these equations (with $X = \mathbb{N}$), as does $\mathrm{id}_{\mathbb{N}}$. By uniqueness, $\iota \circ f = \mathrm{id}_{\mathbb{N}}$, so ι is surjective and $X = \mathbb{N}$, i.e. $\forall n \in \mathbb{N}.\, P(n)$. *Substrate-physically:* the Loop primitive applies the step rule uniformly; if the rule preserves P at every iteration, no iteration breaks P. □

Remark III.7 (Discharge versus closure). Induction is a $\forall n$ claim: it ranges over the type $\mathbb{N}$, not over the finite set of naturals the substrate has so far iterated to. Observation can only confirm $P(0), P(1), \ldots, P(K)$ for some finite K. The universal claim is licensed not by these observations but by the *uniformity* of the substrate's iteration rule (Loop applies the same step at every successor application), combined with the universal property of $\mathbb{N}$ as the initial $(\mathbf{0}, S)$-algebra (Definition III.5). Conservation laws and stable-mode persistence are the empirical discharge that the substrate *has* the iterative structure to which induction applies; they are not the source of the universal quantifier. Whether the closure-to-type-$\mathbb{N}$ step is part of the substrate's primitive content (under Loop plus infinity) or a derivational closure imposed on top of the substrate's actual iterations is a question we treat as part of the framework's metalogic-coextensivity claim (Sec. II A).

C. Two parallel readings of arithmetic

Once the natural numbers exist, the four arithmetic operations admit two parallel substrate-physical readings. The *click-count reading* runs at the level of counted events; the *substrate-amplitude reading* runs at the level of substrate wave-patterns. Both readings are licensed by the framework's commitments and are used together in subsequent derivations. We do not derive their equivalence in this paper; we treat it as a substrate commitment, namely that the substrate's coupling-event structure is the same operational rule at the count-of-events level and at the wave-pattern level, and that the arithmetic produced at either level is therefore the same arithmetic. The corresponding bridge theorem — that the click-count ring of substrate-naturals and the amplitude-level scalar-cross-term ring are isomorphic — is stated in the framework's formal companion [11]; the present paper takes it as given. The empirical pressure on this commitment is that physics uses the same number system for both purposes; a framework that gave them distinct arithmetics would be under immediate empirical strain.

a. Click-count reading.

Definition III.8 (Substrate-addition)**.** *Addition $+ : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ is defined by recursion on the second argument:*

$$m + \mathbf{0} \;=\; m, \tag{27}$$

$$m + S(n) \;=\; S(m + n). \tag{28}$$

The substrate's Loop *executes the recursion: starting from m, apply S exactly n times.*

Definition III.9 (Substrate-multiplication)**.** *Multiplication $\times : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ is defined by recursion on the second argument:*

$$m \times \mathbf{0} \;=\; \mathbf{0}, \tag{29}$$

$$m \times S(n) \;=\; (m \times n) + m. \tag{30}$$

The standard ring identities follow by induction (Theorem III.6).

Lemma III.10 (Associativity of $+$)**.** *For all $a, b, c \in \mathbb{N}$,*

$$(a + b) + c \;=\; a + (b + c). \tag{31}$$

Sketch by induction on c. Base ($c = \mathbf{0}$): $(a + b) + \mathbf{0} = a + b = a + (b + \mathbf{0})$ by Eq. (27). *Step*: assume $(a + b) + c = a + (b + c)$; then

$$(a+b)+S(c) = S((a+b)+c) = S(a+(b+c)) = a+S(b+c) = a+(b+S(c \tag{32}$$

using Eq. (28) repeatedly. □

Lemma III.11 (Left identity for +). $\mathbf{0} + a = a$ *for all* $a \in \mathbb{N}$.

Proof. By induction on a. *Base*: $\mathbf{0} + \mathbf{0} = \mathbf{0}$ by Eq. (27). *Step*: assuming $\mathbf{0} + a = a$, $\mathbf{0} + S(a) \overset{(28)}{=} S(\mathbf{0} + a) = S(a)$. □

Lemma III.12 (Successor on the left). $S(a)+b = S(a+b)$ *for all* $a, b \in \mathbb{N}$.

Proof. By induction on b. *Base*: $S(a) + \mathbf{0} = S(a) = S(a+\mathbf{0})$ by Eq. (27). *Step*: assuming $S(a)+b = S(a+b)$, $S(a) + S(b) \overset{(28)}{=} S(S(a) + b) = S(S(a + b)) \overset{(28)}{=} S(a + S(b))$. □

Lemma III.13 (Commutativity of +). $a+b = b+a$ *for all* $a, b \in \mathbb{N}$.

Proof. By induction on b. *Base*: $a+\mathbf{0} = a \overset{\text{Lem. III.11}}{=} \mathbf{0}+a$. *Step*: $a + S(b) \overset{(28)}{=} S(a + b) = S(b + a) \overset{\text{Lem. III.12}}{=} S(b) + a$. □

Lemma III.14 (Left-cancellation in $\mathbb{N}$). $a+b = a+c \Rightarrow b = c$ *for all* $a, b, c \in \mathbb{N}$.

Proof. By induction on a. *Base*: $\mathbf{0} + b = \mathbf{0} + c$ gives $b = c$ by Lem. III.11. *Step*: $S(a) + b = S(a) + c$ gives $S(a + b) = S(a + c)$ by Lem. III.12, hence $a + b = a + c$ by Ax. III.3, hence $b = c$ by induction hypothesis. □

Lemma III.15 (Distributivity). $a \times (b + c) = (a \times b) + (a \times c)$ *for all* $a, b, c \in \mathbb{N}$.

Proof. By induction on c. *Base*: $a \times (b + \mathbf{0}) = a \times b = (a \times b) + \mathbf{0} = (a \times b) + (a \times \mathbf{0})$ using Eqs. (27) and (29). *Step*:

$$\begin{aligned} a \times (b + S(c)) &\overset{(28)}{=} a \times S(b + c) \\ &\overset{(30)}{=} (a \times (b + c)) + a \\ &= ((a \times b) + (a \times c)) + a \\ &\overset{\text{Lem. III.10}}{=} (a \times b) + ((a \times c) + a) \\ &\overset{(30)}{=} (a \times b) + (a \times S(c)). \end{aligned}$$

□

Lemma III.16 (Associativity and commutativity of ×). $(a\times b)\times c = a\times(b\times c)$ *and* $a\times b = b\times a$ *for all* $a, b, c \in \mathbb{N}$.

Sketch. Standard inductions: associativity by induction on c using Lem. III.15; commutativity by induction on b, using Lem. III.15 and the analogue $S(a) \times b = (a \times b) + b$ (itself proved by induction on b). □

Theorem III.17 ($\mathbb{N}$ is a commutative semiring). $(\mathbb{N}, +, \times, \mathbf{0}, \mathbf{1})$ *with* $\mathbf{1} := S(\mathbf{0})$ *is a commutative semiring: addition is associative (Lem. III.10) and commutative (Lem. III.13) with identity* $\mathbf{0}$ *(Lem. III.11); multiplication is associative and commutative (Lem. III.16) with identity* $\mathbf{1}$*: the right identity* $a \times \mathbf{1} \overset{(30)}{=} (a \times \mathbf{0}) + a \overset{(29)}{=} \mathbf{0} + a \overset{\text{Lem. III.11}}{=} a$*, and the left identity* $\mathbf{1} \times a = a \times \mathbf{1} = a$ *by commutativity (Lem. III.16); distributivity holds (Lem. III.15).*

b. Substrate-physical reading. + is the substrate iterating successor under Loop: a detector at click-count m that registers n more clicks ends up at $m+n$. × is the substrate iterating addition under Loop: a detector observed across n measurement windows, each registering m clicks, has cumulative count $m\times n$. Associativity, commutativity, and distributivity are not extra postulates — they follow from the recursive definitions Eqs. (27)–(30), which themselves record what Loop does mechanically.

c. Substrate-amplitude reading. At the substrate-amplitude level, the same four operations have direct substrate-physical content as operations on substrate wave-pattern amplitudes:

- *Addition is linear superposition.* When two substrate-amplitude patterns occupy the same region of substrate position-space, the substrate's free-propagation regime (the linear sector of its elastic response) commits that the resulting amplitude is the sum of the participating amplitudes. The substrate-physical content of addition at the amplitude level is therefore the substrate's linear superposition of two amplitude patterns.

- *Subtraction is destructive interference.* Inverse of linear superposition: removing a participating amplitude from a superposed configuration reduces the resulting amplitude by the removed amplitude's content. This is observable in the destructive interference of substrate-amplitude patterns at anti-aligned phase, where the resulting amplitude is the difference between the participating amplitudes.

- *Multiplication captures cross-term coupling.* When two substrate-amplitude patterns overlap in the substrate's nonlinear regime, the cross-term in the substrate's nonlinear response is bilinear in the participating amplitudes: doubling either input doubles the cross-term contribution; the cross-term vanishes when either input vanishes. We do not claim multiplication is the *only* mathematical structure that exhibits this bilinearity (Lie brackets, tensor products, and associative non-commutative products are all bilinear), but multiplication on $\mathbb{N}$ — commutative, associative, distributive over addition — is what the substrate's leading-order isotropic cross-term has when the participating modes are scalar amplitudes. The substrate-physical reading of multiplication at the

amplitude level is therefore this bilinear cross-term in its scalar, commutative special case; more general cross-coupling structures (non-commutative, anti- symmetric, higher-rank) require the corresponding generalisations of multiplication and emerge in later sections.

- *Division is the formal inverse, with a remainder.* Inverse of cross-term coupling: separating a cross-term-coupled configuration back into its participating modes leaves a remainder consisting of the cross-term substrate-tension content that does not decompose cleanly into either participating mode individually. This remainder is what mainstream physics measures as the *binding energy* of bound configurations. We do not claim the remainder is itself the gauge-boson content (e.g. gluons in hadrons); the gauge bosons are independently characterised substrate-Lie-group connections, and the relation between the algebraic division remainder and the chromodynamic field configurations sourcing hadronic binding is non-trivial. The substrate-physical content of division at the amplitude level is the inverse cross-term operation, leaving binding-energy substrate tension as the unmodelled remainder.

Two technical caveats. First, both the click-count and substrate- amplitude readings invoke different physical regimes for $+$ and $\times$: the click-count addition uses the substrate's iteration primitive, and the amplitude addition uses the substrate's *linear* free-propagation regime, while the amplitude multiplication uses the substrate's *nonlinear* cross-term regime. The resulting arithmetic comes out the same on both readings (commutative ring with addition and multiplication), but the dynamical regimes are distinct. Second, subtraction on $\mathbb{N}$ is only partially defined ($3-5$ is not a natural number); the destructive-interference reading inherits this partiality, and the total subtraction operation requires the integers, constructed in Sec. IV.

IV. FROM ARITHMETIC TO THE STANDARD NUMBER SYSTEMS

A. The integers

The substrate-physical pressure: opposing-tally processes (electric charge accumulators, account balances, particle counters with addition and removal). The construction is the standard Grothendieck quotient.

Definition IV.1 (Substrate-integers)**.** *On* $\mathbb{N}\times\mathbb{N}$ *define the relation*

$$(a,b)\sim_{\mathbb{Z}}(c,d) \iff a+d=c+b. \tag{33}$$

Lemma IV.2. *The relation in Eq.* (33) *is an equivalence relation on* $\mathbb{N}\times\mathbb{N}$*.*

Proof. Reflexivity: $a+b=a+b$. *Symmetry*: $a+d=c+b \Rightarrow c+b=a+d$ trivially. *Transitivity*: from $(a,b)\sim_{\mathbb{Z}}(c,d)$ and $(c,d)\sim_{\mathbb{Z}}(e,f)$ we have $a+d=c+b$ and $c+f=e+d$. Adding the two equations and rearranging via associativity and commutativity of $+$ (Lems. III.10, III.13),

$$(a+f)+(c+d) \;=\; (e+b)+(c+d). \tag{34}$$

By left-cancellation in $\mathbb{N}$ (Lem. III.14), $a+f=e+b$, i.e. $(a,b)\sim_{\mathbb{Z}}(e,f)$. □

Definition IV.3. $\mathbb{Z}:=(\mathbb{N}\times\mathbb{N})/\sim_{\mathbb{Z}}$. *Write* $[a,b]$ *for the equivalence class of* (a,b)*, so* $[a,b]$ *represents* $a-b$*.*

Definition IV.4 (Integer operations)**.**

$$[a,b]+[c,d] \;:=\; [a+c,\; b+d], \tag{35}$$
$$-[a,b] \;:=\; [b,a], \tag{36}$$
$$[a,b]\times[c,d] \;:=\; [ac+bd,\; ad+bc]. \tag{37}$$

Theorem IV.5. $(\mathbb{Z},+,\times,[\mathbf{0},\mathbf{0}],[\mathbf{1},\mathbf{0}])$ *is a commutative ring. The map* $\iota:\mathbb{N}\to\mathbb{Z}$ *sending* $n\mapsto[n,\mathbf{0}]$ *is an injective semiring homomorphism; it embeds* $\mathbb{N}$ *into* $\mathbb{Z}$ *as the non-negative integers.*

Proof. Well-definedness of $+,\times,-$ on equivalence classes is checked by substituting $\sim_{\mathbb{Z}}$-equivalent representatives and using Theorem III.17 (commutative-semiring identities in $\mathbb{N}$). Associativity, commutativity, and distributivity in $\mathbb{Z}$ then lift component-wise from the corresponding identities in $\mathbb{N}$. The additive inverse:

$$[a,b]+(-[a,b])=[a,b]+[b,a]=[a+b,\; b+a]. \tag{38}$$

We must check $(a+b,b+a)\sim_{\mathbb{Z}}(\mathbf{0},\mathbf{0})$, which by Eq. (33) is the equation $(a+b)+\mathbf{0}=\mathbf{0}+(b+a)$. The left side equals $a+b$ by Eq. (27); the right side equals $b+a=a+b$ by Lem. III.11 and Lem. III.13. Hence $[a,b]+(-[a,b])=[\mathbf{0},\mathbf{0}]$, so $-[a,b]$ is the additive inverse of $[a,b]$. Injectivity of $\iota:n\mapsto[n,\mathbf{0}]$ follows from left-cancellation in $\mathbb{N}$ (Lem. III.14): $[m,\mathbf{0}]=[n,\mathbf{0}]$ gives $m+\mathbf{0}=n+\mathbf{0}$, i.e. $m=n$. □

The discharge witness is the existence of substrate processes whose tally values include both signs; the construction $\mathbb{Z}=(\mathbb{N}\times\mathbb{N})/\sim$ is a closure step performed in the framework's metalogic on the discharged $\mathbb{N}$.

B. The rationals

Definition IV.6 (Substrate-rationals)**.** *On* $\mathbb{Z}\times(\mathbb{Z}\setminus\{0\})$ *define*

$$(p,q)\sim_{\mathbb{Q}}(r,s) \iff ps=rq, \tag{39}$$

and let $\mathbb{Q}:=(\mathbb{Z}\times(\mathbb{Z}\setminus\{0\}))/\sim_{\mathbb{Q}}$*. Write* $[p,q]_{\mathbb{Q}}$ *or* $\frac{p}{q}$ *for the equivalence class of* (p,q)*.*

Definition IV.7 (Rational operations)**.**

$$\frac{p}{q} + \frac{r}{s} := \frac{ps+rq}{qs}, \tag{40}$$
$$\frac{p}{q} \times \frac{r}{s} := \frac{pr}{qs}, \tag{41}$$
$$\left(\frac{p}{q}\right)^{-1} := \frac{q}{p} \quad (p \neq 0). \tag{42}$$

Theorem IV.8. $(\mathbb{Q}, +, \times)$ *is a field of characteristic* 0*. The map* $\jmath : \mathbb{Z} \to \mathbb{Q}$ *sending* $n \mapsto \frac{n}{1}$ *is an injective ring homomorphism.*

Sketch. $\sim_{\mathbb{Q}}$ is an equivalence relation by an argument parallel to Lemma IV.2. The operations Eqs. (40)–(41) are well-defined on equivalence classes (checked by substituting equivalent representatives and using ring identities in $\mathbb{Z}$). Field axioms hold by direct computation; in particular every non-zero element has the inverse given. Characteristic zero follows from $\jmath$'s injectivity on $\mathbb{Z}$. □

The substrate-physical content of a rational is a *ratio of two counted integer-valued substrate processes*: the number of decays in channel A over the total number of decays in a sample (a count over a count), the count of atomic-clock cycles per second-marker, the lattice index of a stable mode within a multiplet. General real-valued ratios — branching fractions, transition frequency ratios, mixing angles — are not themselves rationals; they are real numbers approximated to arbitrary precision by Cauchy sequences of rationals, and live in the substrate-reals constructed in Sec. IV C.

C. The reals

Definition IV.9 (Cauchy sequence in $\mathbb{Q}$)**.** *A sequence* $(a_n)_{n\in\mathbb{N}}$ *in* $\mathbb{Q}$ *is* Cauchy *if*

$$\forall \varepsilon \in \mathbb{Q}_{>0}\, \exists N \in \mathbb{N}\, \forall m, n \geq N. \quad |a_m - a_n| < \varepsilon. \tag{43}$$

Definition IV.10 (Cauchy equivalence)**.** *For Cauchy sequences* $(a_n), (b_n)$ *in* $\mathbb{Q}$*,*

$$(a_n) \sim_{\mathbb{R}} (b_n) \iff \forall \varepsilon \in \mathbb{Q}_{>0}\, \exists N\, \forall n \geq N.\, |a_n - b_n| < \varepsilon. \tag{44}$$

Definition IV.11 (Substrate-reals)**.** $\mathbb{R} := \mathcal{C}(\mathbb{Q})/\sim_{\mathbb{R}}$*, where* $\mathcal{C}(\mathbb{Q})$ *is the set of Cauchy sequences in* $\mathbb{Q}$*.*

Definition IV.12 (Real operations)**.** *For* $[(a_n)], [(b_n)] \in \mathbb{R}$*:*

$$[(a_n)] + [(b_n)] := [(a_n + b_n)], \tag{45}$$
$$[(a_n)] \times [(b_n)] := [(a_n \cdot b_n)]. \tag{46}$$

Order: $[(a_n)] < [(b_n)]$ *iff* $\exists \varepsilon > 0\, \exists N\, \forall n \geq N.\, b_n - a_n > \varepsilon$*.*

Theorem IV.13 ($\mathbb{R}$ is a complete ordered field)**.** $(\mathbb{R}, +, \times, <)$ *is a totally ordered field, and every Cauchy sequence in* $\mathbb{R}$ *converges to a unique element of* $\mathbb{R}$*.*

Sketch. Well-definedness of $+, \times$*.* Cauchy sequences are bounded ($|a_n| \leq M$ for some $M \in \mathbb{Q}$, since for $\varepsilon = 1$ there is N with $|a_n| \leq |a_N| + 1$ for $n \geq N$). For sums, $|(a_m + b_m) - (a_n + b_n)| \leq |a_m - a_n| + |b_m - b_n|$, Cauchy again. For products, with $|a_n|, |b_n| \leq M$, $|a_m b_m - a_n b_n| \leq M(|a_m - a_n| + |b_m - b_n|)$, also Cauchy. The operations preserve $\sim_{\mathbb{R}}$ by an analogous $\varepsilon/(2M)$ estimate. *Field axioms.* Associativity, commutativity, distributivity, and the identities $0 = [(\mathbf{0})_n]$, $1 = [(\mathbf{1})_n]$ lift from $\mathbb{Q}$. The non-trivial part is the multiplicative inverse: if $[(a_n)] \neq 0$, then $(a_n) \not\sim_{\mathbb{R}} (\mathbf{0})_n$, so there is $\delta \in \mathbb{Q}_{>0}$ with $|a_n| \geq \delta$ for all sufficiently large n (otherwise a_n would have a subsequence $\to 0$, contradicting Cauchy non-equivalence to 0). Replacing finitely many initial terms by δ gives a representative bounded below; then $1/a_n$ is defined for all n and Cauchy because $|1/a_m - 1/a_n| = |a_n - a_m|/(|a_m||a_n|) \leq |a_m - a_n|/\delta^2$. *Completeness.* Given a Cauchy sequence $([(a_n^{(k)})])_k$ in $\mathbb{R}$, choose representatives $(a_n^{(k)})$ and pick N_k large enough that $|a_n^{(k)} - a_{N_k}^{(k)}| < 2^{-k}$ for $n \geq N_k$. Set $d_k := a_{N_k}^{(k)}$. Then (d_k) is Cauchy in $\mathbb{Q}$ (by combining the diagonal bound with the Cauchy hypothesis on the equivalence classes), and $[(d_k)] \in \mathbb{R}$ is the limit of the original sequence. □

Remark IV.14 (Discharge versus closure, again). Observation supplies the rational *members* of any such Cauchy sequence (each refined measurement is a rational with finite precision); it does not, on its own, supply the closure-to-limit step. The substrate-real is the *equivalence class*, and packaging that class as a single object is a derivational closure step performed on top of the substrate-rationals, licensed by the substrate's commitment to Loop (every Cauchy sequence is generated by an unbounded iteration) plus extensionality (two iterations producing the same limit-witnessing sequence are the same closure object). The real numbers are uncountable while observations are countable: the construction does not claim every real has been measured, only that every measured quantity that admits indefinite refinement is representable as some Cauchy class.

D. Sets

Definition IV.15 (Substrate-type universe)**.** *The substrate's type-formation primitive produces a cumulative sequence of universes*

$$\mathcal{U}_0 \subset \mathcal{U}_1 \subset \mathcal{U}_2 \subset \cdots \tag{47}$$

where $\mathcal{U}_0$ *contains substrate configurations and each* $\mathcal{U}_{i+1}$ *contains substrate-sets whose defining predicates range over* $\mathcal{U}_i$*.*

Definition IV.16 (Substrate-set)**.** *A* substrate-set at level $i+1$ *is determined by a predicate* $P : \mathcal{U}_i \to \{\bot, \top\}$*:*

$$\{x \in \mathcal{U}_i : P(x)\} \in \mathcal{U}_{i+1}. \tag{48}$$

Predicates ranging over $\mathcal{U}_i$ produce sets at level $i+1$, not at level i or below; this rules out unrestricted comprehension.

Proposition IV.17 (Russell's paradox is excluded)**.** *The expression $\{x : x \notin x\}$ is not a well-formed substrate-set. By Definition IV.15, an element x lives at some level $\mathcal{U}_i$, and the substrate-sets that take x as a member are at level $\mathcal{U}_{i+1}$ — their defining predicate ranges over $\mathcal{U}_i$. Asking whether $x \in x$ would require x to be both at level $\mathcal{U}_i$ (as the candidate member) and at level $\mathcal{U}_{i+1}$ (as the containing substrate-set) at the same time, which is type-incoherent. The type discipline makes the predicate ill-typed.*

Definition IV.18 (Set operations)**.** *For substrate-sets $A = \{x : P(x)\}$ and $B = \{x : Q(x)\}$ at the same type level:*

$$A \cup B := \{x : P(x) \vee Q(x)\}, \tag{49}$$

$$A \cap B := \{x : P(x) \wedge Q(x)\}, \tag{50}$$

$$A^c := \{x : \neg P(x)\}, \tag{51}$$

$$A \subseteq B :\Longleftrightarrow \forall x.\, P(x) \to Q(x). \tag{52}$$

The substrate's And/Or/Not-coupling primitives directly discharge Eqs. (49)–(51); physics taxonomies routinely combine sets this way (hadrons $\cup$ leptons = fermions; charged $\cap$ stable = stable charged species; complement of charged within all = neutral).

E. Topologies and continuity

Definition IV.19 (Substrate-topology)**.** *A* topology *on a substrate-set X is a collection $\mathcal{T} \subseteq \mathcal{P}(X)$ of subsets of X (the* open sets*) satisfying:*

$$\textit{(T1)} \quad \emptyset \in \mathcal{T}, \quad X \in \mathcal{T}. \tag{53}$$

$$\textit{(T2)} \quad \forall \{U_\lambda\}_{\lambda\in\Lambda} \subseteq \mathcal{T}. \quad \bigcup_{\lambda\in\Lambda} U_\lambda \in \mathcal{T}. \tag{54}$$

$$\textit{(T3)} \quad \forall U_1, \ldots, U_n \in \mathcal{T}. \quad \bigcap_{i=1}^{n} U_i \in \mathcal{T}. \tag{55}$$

The pair $(X, \mathcal{T})$ is a topological space*.*

Definition IV.20 (Continuous map)**.** *A map $f : (X, \mathcal{T}_X) \to (Y, \mathcal{T}_Y)$ is* continuous *if*

$$\forall V \in \mathcal{T}_Y. \quad f^{-1}(V) \in \mathcal{T}_X. \tag{56}$$

The substrate-physical reading: a measurement-precision band, a position-uncertainty region, and a frequency-window each correspond to an open set in the relevant topology; And/Or/Not-coupling on neighbourhood predicates discharges the closure rules Eqs. (54)–(55) via Definition IV.18. Two qualifications carry over from the substrate-real construction: (i) laboratory measurements furnish only finitely many concrete open sets, while $\mathcal{T}_{\mathbb{R}^n}$ has uncountably many; the discharge covers the closure rules, not the totality of open sets; (ii) the closure to "arbitrary unions, finite intersections" is a derivational step on top of the observed neighbourhood structure.

Proposition IV.21 (Continuity preserves measurement-discriminability)**.** *Every measurement system gives a continuous map from input configurations to output values, and conservation laws give continuous maps from initial states to final states.*

The substrate-physical content of Proposition IV.21: without continuity, no input precision could guarantee any output precision; physical prediction would be impossible. The empirical witness is that prediction *is* possible.

V. FROM TOPOLOGY TO GEOMETRY

A. Smooth manifolds

Definition V.1 (Chart and atlas)**.** *Let M be a Hausdorff, second-countable topological space. A* chart *on M is a pair (U, φ) where $U \subseteq M$ is open and $\varphi : U \to \varphi(U) \subseteq \mathbb{R}^n$ is a homeomorphism onto an open subset of $\mathbb{R}^n$. Two charts $(U, \varphi), (V, \psi)$ are C^∞-*compatible *if either $U \cap V = \emptyset$ or the transition map*

$$\psi \circ \varphi^{-1} : \varphi(U \cap V) \to \psi(U \cap V) \tag{57}$$

is a C^∞ diffeomorphism between open subsets of $\mathbb{R}^n$. A C^∞ atlas *on M is a collection of pairwise-compatible charts whose domains cover M.*

Definition V.2 (Smooth manifold)**.** *A* smooth n-manifold *is a Hausdorff, second-countable space M together with a maximal C^∞ atlas of charts to $\mathbb{R}^n$.*

Definition V.3 (Tangent space)**.** *The* tangent space *T_pM at $p \in M$ is the $\mathbb{R}$-vector space of equivalence classes of smooth curves $\gamma : (-\varepsilon, \varepsilon) \to M$ with $\gamma(0) = p$, where $\gamma_1 \sim \gamma_2$ iff $(d/dt)|_{t=0}(\varphi\circ\gamma_1) = (d/dt)|_{t=0}(\varphi\circ\gamma_2)$ in some chart (U, φ) with $p \in U$ (the condition is independent of the chart by the chain rule and the C^∞ compatibility of transitions).*

Lemma V.4 (Chain rule for transitions)**.** *For two charts (U, φ) and (V, ψ) with $p \in U \cap V$ and smooth curve γ,*

$$\left.\frac{d}{dt}\right|_0 (\psi \circ \gamma) = D(\psi \circ \varphi^{-1})\big(\varphi(p)\big) \cdot \left.\frac{d}{dt}\right|_0 (\varphi \circ \gamma). \tag{58}$$

The substrate-physical reading: the substrate's elastic response is locally smooth (small displacements produce smoothly varying restoring tensions, not discontinuous jumps); particle worldlines are differentiable except at scattering events; the CMB temperature field is smooth at every resolvable angular scale. A substrate-tangent vector at a configuration is the directional rate-of-change of any substrate-observable along an infinitesimal substrate-displacement — drift velocities, pressure gradients, field strengths, gauge-potential gradients are all instances of Definition V.3.

B. Tensor fields

Definition V.5 (Tensor at a point)**.** *The* cotangent space *at $p \in M$ is the dual vector space $T_p^*M = \mathrm{Hom}_{\mathbb{R}}(T_pM, \mathbb{R})$. A (p,q)-*tensor at p *is a multilinear map*

$$T : \underbrace{T_p^*M \times \cdots \times T_p^*M}_{p\ copies} \times \underbrace{T_pM \times \cdots \times T_pM}_{q\ copies} \longrightarrow \mathbb{R}. \tag{59}$$

The space of (p,q)-tensors at p is denoted $T_p^{(p,q)}M = (T_pM)^{\otimes p} \otimes (T_p^*M)^{\otimes q}$. A tensor field *of type (p,q) on M is a smooth assignment $M \ni p \mapsto T(p) \in T_p^{(p,q)}M$.*

Definition V.6 (Multilinearity)**.** *T is multilinear iff for each slot i and scalars $\alpha, \beta \in \mathbb{R}$,*

$$T(\ldots, \alpha v + \beta w, \ldots) = \alpha\, T(\ldots, v, \ldots) + \beta\, T(\ldots, w, \ldots). \tag{60}$$

Theorem V.7 (Components under chart transitions)**.** *Under a change of chart $\psi \circ \varphi^{-1}$ with Jacobian $J^a{}_b = \partial\psi^a/\partial\varphi^b$, the components of a (p,q)-tensor transform as*

$$T'^{a_1\cdots a_p}{}_{b_1\cdots b_q} = J^{a_1}{}_{c_1} \cdots J^{a_p}{}_{c_p} (J^{-1})^{d_1}{}_{b_1} \cdots (J^{-1})^{d_q}{}_{b_q}\, T^{c_1\cdots c_p}{}_{d_1\cdots d_q}. \tag{61}$$

The substrate-physical reading: the elastic response couples several directions multilinearly; doubling the input strain in any one direction doubles the output stress in each output direction (linearity in each slot, Eq. (60)); adding two input strains adds the two output stress responses (the second linearity property). The stress–energy tensor $T_{\mu\nu}$ is a $(0,2)$-tensor field, the Riemann curvature tensor $R^\rho{}_{\sigma\mu\nu}$ is a $(1,3)$-tensor field, and Eq. (61) is what makes "component-of-stress-in-this-direction" a substrate-physically meaningful quantity rather than a chart-dependent artefact.

C. Differential forms and exterior calculus

Definition V.8 (Differential k-form)**.** *A k-form on M is a $(0,k)$-tensor field $\omega : (TM)^{\otimes k} \to \mathbb{R}$ that is alternating:*

$$\omega(v_{\sigma(1)}, \ldots, v_{\sigma(k)}) = \mathrm{sgn}(\sigma)\, \omega(v_1, \ldots, v_k) \tag{62}$$

for every permutation $\sigma \in S_k$. The space of k-forms is denoted $\Omega^k(M)$; $\Omega^0(M) = C^\infty(M, \mathbb{R})$.

Definition V.9 (Exterior derivative)**.** *There is a unique linear operator $d : \Omega^k(M) \to \Omega^{k+1}(M)$ for $k \geq 0$ satisfying:*

$$\textit{(D1)} \quad df = \partial_i f\, dx^i \quad \textit{for } f \in \Omega^0(M), \tag{63}$$

$$\textit{(D2)} \quad d(\omega \wedge \eta) = d\omega \wedge \eta + (-1)^{|\omega|}\, \omega \wedge d\eta, \tag{64}$$

$$\textit{(D3)} \quad d \circ d = 0. \tag{65}$$

In coordinates, for $\omega = \frac{1}{k!}\, \omega_{i_1\cdots i_k}\, dx^{i_1} \wedge \cdots \wedge dx^{i_k}$,

$$d\omega = \tfrac{1}{k!}\, \partial_j \omega_{i_1\cdots i_k}\, dx^j \wedge dx^{i_1} \wedge \cdots \wedge dx^{i_k}. \tag{66}$$

Theorem V.10 ($d^2 = 0$)**.** *For every $\omega \in \Omega^k(M)$, $d(d\omega) = 0$.*

Proof. On a 0-form f, $df = \partial_i f\, dx^i$, and

$$d(df) = \partial_j\partial_i f\, dx^j \wedge dx^i = \tfrac{1}{2}(\partial_j\partial_i - \partial_i\partial_j) f\, dx^j \wedge dx^i = 0 \tag{67}$$

by Schwarz's theorem ($\partial_i\partial_j f = \partial_j\partial_i f$ for $f \in C^2$; the smooth-manifold structure of Definition V.2 guarantees C^∞ regularity locally) combined with the antisymmetry of $\wedge$. The general case follows by linearity and the Leibniz rule (D2). □

Theorem V.11 (Stokes)**.** *For $\omega \in \Omega^{k-1}(M)$ with compact support and an oriented k-dimensional submanifold-with-boundary $\Omega \subseteq M$,*

$$\int_\Omega d\omega \;=\; \int_{\partial\Omega} \omega. \tag{68}$$

Corollary V.12 (Algebraic and topological identities)**.** *$d^2 = 0$ on forms and $\partial^2 = \emptyset$ on chains are dual statements: applying d to $\omega \in \Omega^{k-2}$ and integrating over $\partial(\partial\Omega)$ gives, by Stokes applied twice,*

$$0 = \int_{\partial(\partial\Omega)} \omega = \int_{\partial\Omega} d\omega = \int_\Omega d^2\omega = 0, \tag{69}$$

consistent for every ω iff both $\partial^2\Omega = \emptyset$ and *$d^2\omega = 0$.*

Theorem V.13 (Maxwell's homogeneous equations)**.** *Let $A \in \Omega^1(M)$ be the substrate's electromagnetic gauge potential and $F := dA \in \Omega^2(M)$ the field strength. Then*

$$dF = d(dA) = 0 \tag{70}$$

by Theorem V.10, which on a topologically trivial 4-region is equivalent to the no-monopole equation $\nabla \cdot \mathbf{B} = 0$ and Faraday's law $\nabla \times \mathbf{E} + \partial_t \mathbf{B} = 0$. On a region with $H^2(M;\mathbb{Z}) \neq 0$, $F/(2\pi)$ may be closed but not exact (with respect to integer cohomology), and its class $[F/(2\pi)] \in H^2(M;\mathbb{Z})$ is the Dirac-quantised magnetic-monopole charge; writing $F = dA$ globally is precisely the assertion that this class vanishes.

Substrate-physical reading. The antisymmetry Eq. (62) captures the observation that oriented substrate quantities (area swept, flux through a surface, vortex circulation, neutrino helicity, parity-violating weak amplitudes) change sign under orientation reversal. Theorem V.10 together with Stokes' theorem give every observed conservation law as the integral of a closed differential form: charge, energy, momentum, angular momentum, baryon number, and lepton number conservation are each $\int_{\partial\Omega} \omega = \int_\Omega d\omega = 0$ for the appropriate ω.

D. Vector calculus and fluid dynamics

Vector calculus on Euclidean $\mathbb{R}^3$ is the special case of exterior calculus where the substrate's flat metric provides the musical isomorphisms $\flat : TM \to T^*M$ and

$\sharp : T^*M \to TM$, plus the Hodge star $\star : \Omega^k \to \Omega^{n-k}$. With $n = 3$:

$$\nabla f \;=\; (df)^\sharp, \quad f \in \Omega^0, \tag{71}$$

$$\nabla \times \mathbf{A} \;=\; \big(\star\, d(\mathbf{A}^\flat)\big)^\sharp, \tag{72}$$

$$\nabla \cdot \mathbf{A} \;=\; \star\, d \star (\mathbf{A}^\flat). \tag{73}$$

The exterior-derivative chain $\Omega^0 \xrightarrow{d} \Omega^1 \xrightarrow{d} \Omega^2 \xrightarrow{d} \Omega^3$ in $\mathbb{R}^3$ becomes the vector-calculus chain $f \xrightarrow{\nabla} \mathbf{F} \xrightarrow{\nabla\times} \mathbf{G} \xrightarrow{\nabla\cdot} h$, with $d^2 = 0$ (Theorem V.10) translating into the identities

$$\nabla \times (\nabla f) = \mathbf{0}, \qquad \nabla \cdot (\nabla \times \mathbf{A}) = 0. \tag{74}$$

Theorem V.14 (Continuity equation)**.** *For a substrate-conserved density ρ and current $\mathbf{j}$,*

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{j} \;=\; 0 \tag{75}$$

is exactly Stokes' theorem (Theorem V.11) applied to the 3-form $J = \rho\, dx \wedge dy \wedge dz - (\star_3 \mathbf{j}^\flat) \wedge dt$ on a 4-volume bounded by spatial regions at t_1, t_2, where $\star_3$ is the 3-dimensional Hodge star (so $\star_3 \mathbf{j}^\flat$ is the 2-form $j^x dy \wedge dz + j^y dz \wedge dx + j^z dx \wedge dy$): $\int_{V\times\{t_2\}} \rho - \int_{V\times\{t_1\}} \rho = -\int_{\partial V \times [t_1,t_2]} \mathbf{j}^\flat$, the integrated form of Eq. (75)*.*

The Euler and Navier–Stokes equations follow by augmenting Eq. (75) with momentum-flux and viscous-stress contributions in the standard way; each is the substrate's local "no spontaneous creation or destruction of conserved quantities" applied to a different conserved quantity (mass, momentum, energy).

VI. FROM GEOMETRY TO SYMMETRY

A. Lie groups

Definition VI.1 (Lie group)**.** *A* Lie group *is a smooth manifold G (Definition V.2) equipped with smooth maps*

$$\mu : G \times G \to G, \quad \mu(g,h) = g \cdot h, \tag{76}$$

$$\iota : G \to G, \quad \iota(g) = g^{-1}, \tag{77}$$

and a distinguished element $e \in G$, satisfying the group axioms:

$$\mu(\mu(g,h),k) = \mu(g,\mu(h,k)), \quad \mu(g,e) = \mu(e,g) = g,$$
$$\mu(g,\iota(g)) = \mu(\iota(g),g) = e. \tag{78}$$

The substrate-physical process: substrate standing-wave configurations are observed to remain physically equivalent under continuous deformations of certain kinds. Rotating the laboratory frame does not change atomic spectra; performing a global gauge phase rotation on a wavefunction does not change probability densities; an isospin rotation does not change strong-interaction cross-sections within an isospin multiplet. Each is a substrate-Lie-group acting smoothly on substrate configurations.

B. Lie algebras

Definition VI.2 (Lie algebra of a Lie group)**.** *The* Lie algebra *of a Lie group G is the tangent space $\mathfrak{g} = T_eG$ at the identity. For $X \in \mathfrak{g}$, the* left-invariant extension *is the vector field X^L on G defined by*

$$X^L_g \;:=\; (L_g)_*(X), \quad \text{where } L_g(h) = g \cdot h, \tag{79}$$

i.e. X^L_g is the pushforward of X at e along left translation by g. The bracket on $\mathfrak{g}$ is

$$[X,Y] \;:=\; (\mathcal{L}_{X^L} Y^L)_e, \tag{80}$$

the value at e of the Lie derivative of Y^L along X^L. Equivalently, in coordinates near the identity, $[X,Y]$ is the order-XY correction in the Baker–Campbell–Hausdorff expansion of the group multiplication (Theorem VI.4).

Theorem VI.3 (Lie-algebra axioms)**.** *The bracket $[\cdot,\cdot]$ is bilinear, antisymmetric, and satisfies the Jacobi identity:*

$$[\alpha X + \beta Y, Z] = \alpha[X,Z] + \beta[Y,Z], \tag{81}$$

$$[X,Y] = -[Y,X], \tag{82}$$

$$[X,[Y,Z]] + [Y,[Z,X]] + [Z,[X,Y]] = 0. \tag{83}$$

Theorem VI.4 (Exponential map)**.** *There is a smooth map $\exp : \mathfrak{g} \to G$ characterised by the property that $t \mapsto \exp(tX)$ is the integral curve through e of the left-invariant vector field generated by X. Locally near $0 \in \mathfrak{g}$, $\exp$ is a diffeomorphism onto a neighbourhood of $e \in G$. The Baker–Campbell–Hausdorff formula*

$$\exp(X)\exp(Y) = \exp\Big(X+Y+\tfrac{1}{2}[X,Y]+\tfrac{1}{12}\big([X,[X,Y]]-[Y,[X,Y]]\big)+\cdots \tag{84}$$

expresses group multiplication near e entirely in Lie-algebra terms; the bracket Eq. (80) *is precisely the order-XY correction.*

The substrate-physical reading: $\mathfrak{g}$ is the space of infinitesimal substrate symmetry generators; the bracket measures the failure of two symmetries to commute. Thomas precession, the $SU(2)$ commutation of angular-momentum components, and the colour-generator commutators are all instances of Eq. (80); the exponential map integrates an infinitesimal symmetry to a finite one (spin precession in a magnetic field, Lorentz boosts, mass-eigenbasis rotations).

C. Root systems

Definition VI.5 (Cartan subalgebra)**.** *For a complex semisimple Lie algebra $\mathfrak{g}_\mathbb{C}$ (the complexification $\mathfrak{g} \otimes_\mathbb{R} \mathbb{C}$ of a real semisimple algebra $\mathfrak{g}$), a* Cartan subalgebra *$\mathfrak{h} \subseteq \mathfrak{g}_\mathbb{C}$ is a maximal abelian subalgebra consisting of* ad-*semisimple elements (those whose adjoint action on $\mathfrak{g}_\mathbb{C}$ is diagonalisable). By a theorem of Cartan, all Cartan subalgebras are conjugate; their common dimension is the* rank *of $\mathfrak{g}$.*

Theorem VI.6 (Cartan decomposition)**.** *For a complex semisimple Lie algebra $\mathfrak{g}_{\mathbb{C}}$ with Cartan subalgebra $\mathfrak{h}$, there is a vector-space decomposition*

$$\mathfrak{g}_{\mathbb{C}} \;=\; \mathfrak{h} \;\oplus\; \bigoplus_{\alpha\in\Phi} \mathfrak{g}_{\alpha}, \tag{85}$$

where $\Phi \subset \mathfrak{h}^* \setminus \{0\}$ is a finite set of roots *and each* root space

$$\mathfrak{g}_{\alpha} = \{X \in \mathfrak{g}_{\mathbb{C}} : [H, X] = \alpha(H)X \,\forall H \in \mathfrak{h}\} \tag{86}$$

is one-dimensional.

Definition VI.7 (Root system)**.** *A* root system *in a real inner-product space $(V, \langle\cdot,\cdot\rangle)$ is a finite set $\Phi \subset V \setminus \{0\}$ satisfying:*

$$\textit{(R1)}\quad \Phi \text{ spans } V. \tag{87}$$

$$\textit{(R2)}\quad \alpha \in \Phi \Rightarrow \mathbb{R}\alpha \cap \Phi = \{\pm\alpha\}. \tag{88}$$

$$\textit{(R3)}\quad \forall \alpha, \beta \in \Phi. \quad s_{\alpha}(\beta) := \beta - \frac{2\langle\alpha,\beta\rangle}{\langle\alpha,\alpha\rangle}\,\alpha \in \Phi. \tag{89}$$

$$\textit{(R4)}\quad \forall \alpha, \beta \in \Phi. \quad \frac{2\langle\alpha,\beta\rangle}{\langle\alpha,\alpha\rangle} \in \mathbb{Z}. \tag{90}$$

The map s_{α} is the reflection through the hyperplane perpendicular to α.

Definition VI.8 (Coroot)**.** *The* coroot *of α is $\alpha^{\vee} := \frac{2\alpha}{\langle\alpha,\alpha\rangle}$, so Eq.* (89) *reads $s_{\alpha}(\beta) = \beta - \langle\beta,\alpha^{\vee}\rangle\,\alpha$ and Eq.* (90) *reads $\langle\beta,\alpha^{\vee}\rangle \in \mathbb{Z}$.*

Theorem VI.9 (Killing–Cartan crystallographic constraint)**.** *For non-proportional $\alpha, \beta \in \Phi$, let $n(\alpha,\beta) := \langle\beta,\alpha^{\vee}\rangle \in \mathbb{Z}$ and θ the angle between α and β. Then*

$$n(\alpha,\beta)\,n(\beta,\alpha) \;=\; \frac{4\langle\alpha,\beta\rangle^2}{\langle\alpha,\alpha\rangle\langle\beta,\beta\rangle} = 4\cos^2\theta \;\in\; \{0,1,2,3\}, \tag{91}$$

and $\theta \in \{30^\circ, 45^\circ, 60^\circ, 90^\circ, 120^\circ, 135^\circ, 150^\circ\}$.

Proof. The product $n(\alpha,\beta)\,n(\beta,\alpha)$ is a non-negative integer (Eq. (90) squared, with non-negativity from the Cauchy–Schwarz inequality $|\langle\alpha,\beta\rangle|^2 \le \langle\alpha,\alpha\rangle\langle\beta,\beta\rangle$ and $<$ strict for non-proportional vectors). It equals $4\cos^2\theta$ by direct computation. The only allowed values are $0, 1, 2, 3$ (equal to $4\cos^2\theta$ for the listed angles); 4 would force $\cos^2\theta = 1$, contradicting non-proportionality. □

Definition VI.10 (Roots vs. weights)**.** *For a representation $\rho : \mathfrak{g} \to \mathrm{End}(V)$, the* weights *of ρ are the eigenvalues $\lambda \in \mathfrak{h}^*$ of $\rho|_{\mathfrak{h}}$ acting on V:*

$$V_{\lambda} = \{v \in V : \rho(H)v = \lambda(H)v \,\forall H \in \mathfrak{h}\}. \tag{92}$$

Roots are the weights of the adjoint representation ad : $\mathfrak{g} \to \mathrm{End}(\mathfrak{g})$ *specifically.*

The substrate's standing-wave catalogue exhibits quantised charges on observed states: integer multiples of $e/3$ for electric charge (quarks at $\pm e/3, \pm 2e/3$ and integer-charge composites at multiples of e), half-integer or integer spin, the three states of the $SU(3)$ fundamental representation for colour, two states of the $SU(2)$ weak-isospin doublet, and so on. These are weights of the substrate-Lie-algebra acting on the representations carried by physical states (Eq. (92)); only the gauge bosons (adjoint representation) carry roots as their charges. The quantisation Eq. (90) forces both roots and weights onto a discrete lattice; the empirical witness is that the substrate's observable charges populate this lattice (the periodic table, the meson–baryon multiplet structure, the Standard-Model gauge-group decompositions). The three matter generations are a separate datum — a multiplicity of replicas of the same charged-fermion representation, not a weight value — fixed in the framework by the Hodge structure of the Schoen quotient [12] rather than by the root/weight lattice.

D. Coxeter and Weyl groups

Definition VI.11 (Weyl group)**.** *The* Weyl group *$W(\Phi)$ of a root system Φ is the finite subgroup of the orthogonal group $O(V)$ generated by the reflections $\{s_{\alpha} : \alpha \in \Phi\}$ defined in Eq.* (89)*.*

Theorem VI.12 (Coxeter presentation)**.** *$W(\Phi)$ admits a presentation by simple reflections $s_1, \ldots, s_r$ (one per simple root) with relations*

$$s_i^2 = 1, \qquad (s_i s_j)^{m_{ij}} = 1 \quad (i \ne j), \tag{93}$$

where $m_{ij} \in \{2, 3, 4, 6\}$ encodes the angle between α_i and α_j via Eq. (91)*.*

Definition VI.13 (Coxeter element)**.** *The* Coxeter element *is $c := s_1 s_2 \cdots s_r$ (product of all simple reflections). Its conjugacy class in $W(\Phi)$ is independent of the chosen ordering.*

Theorem VI.14 (Coxeter spectrum)**.** *Let h be the order of c in $W(\Phi)$ (the* Coxeter number*), and let r be the rank. The eigenvalues of c acting on the Cartan subalgebra are the r complex numbers*

$$\zeta_k \;=\; e^{2\pi i\, m_k/h}, \qquad k = 1, \ldots, r, \tag{94}$$

where the integers $\{m_1, \ldots, m_r\}$ with $1 = m_1 \le m_2 \le \cdots \le m_r = h - 1$ are the exponents *of the algebra. The size of the root system is*

$$|\Phi| \;=\; h \cdot r. \tag{95}$$

For E_8: rank $r = 8$, Coxeter number $h = 30$, exponents $\{1, 7, 11, 13, 17, 19, 23, 29\}$, $|\Phi| = 240$ [10]. The substrate-physical reading: every substrate-root has a negative counterpart (Eq. (88)); reflections through root hyperplanes are exact substrate symmetries; the Coxeter spectrum encodes the discrete substrate-mode hierarchy that drives the framework's mass-spectrum derivation in [12].

E. The empirical anchor: E_8 in $CoNb_2O_6$

The construction so far is structural. It says that a substrate-Lie- algebra exists with a specific root system and a specific Coxeter group. It does not, in itself, say which specific Lie algebra the substrate selects. That selection is empirical, and the selection has been observed.

Coldea *et al.* [7] measured the spectrum of the quasi-one-dimensional Ising ferromagnet $CoNb_2O_6$ near its quantum critical point. The lowest two excitation masses were found to satisfy $m_2/m_1 = \varphi = (1+\sqrt{5})/2$ to part-in-10^3 precision — the golden ratio. This is the leading mass ratio of the Zamolodchikov E_8 spectrum [8]: eight integrable particle masses indexed by the nodes of the E_8 Dynkin diagram, with the mass values fixed (up to overall scale) by the Perron–Frobenius eigenvector of the E_8 Cartan matrix.

The empirical content: E_8 root-system structure is observable in the laboratory at the level of mass ratios in a system that knows nothing about high-energy physics. This is the framework's evidence that the substrate's mode catalogue selects *at least one* copy of the E_8 root system rather than a generic Lie algebra. The further identification of $E_8 \times E_8$ as the substrate's full symmetry algebra is a separate step: the doubling has its own motivation in heterotic-string anomaly cancellation and modular invariance [11, 12], and the $CoNb_2O_6$ observation does not, by itself, force the doubling. We treat the single-E_8 part as discharged by Coldea and the doubling as a separate framework commitment with its own empirical inputs (the cancellation of gauge anomalies in the Standard Model, the visible-vs-dark sector splitting). The bootstrap chain in the present paper runs only as far as "substrate-Lie- algebra with one or more E_8 factors"; the specific selection of the doubled heterotic algebra is established in the companion work.

F. The McKay correspondence and icosahedral seeding

The McKay correspondence [9] establishes a bijection between finite subgroups of $SU(2)$ and the simply-laced ADE Dynkin diagrams via the irreducible representations on resolution divisors of the corresponding du Val singularities. The binary icosahedral group is the McKay dual of E_8. The McKay correspondence is, in itself, a representation-theoretic statement about resolutions of $\mathbb{C}^2/G$ singularities; it does not, on its own, predict mass ratios. The Zamolodchikov E_8 mass spectrum measured by Coldea *et al.* arises from the integrable structure of the perturbed Ising CFT, not from the McKay correspondence directly. The framework's claim is the more modest statement that the McKay-dual binary icosahedral group — and in particular its order-5 rotational substructure — is structurally available as a discrete symmetry whenever the substrate's E_8 acts; the framework then *selects* this order-5 substructure as the harmonic seed of its mass-spectrum construction in the companion paper [12]. The selection is a framework commitment, not a McKay theorem; the empirical content that discharges it is the agreement of the resulting φ-chain predictions with measured fermion masses and with the $CoNb_2O_6$ golden-ratio mass ratio at the appropriate level of the Coxeter spectrum.

VII. THE DISSOLUTION OF WIGNER'S PUZZLE

We can now state the resolution. The argument is short. By *structural identity* we mean specifically: the formal description licensed by the substrate's primitive content, when restricted to the slice of mathematics enumerated in Sec. II A (elementary number theory through Coxeter classification), produces the same objects, up to bridge-theorem isomorphism, as the standard-mathematics objects of those theories. This is an isomorphism claim between two presentations, not a metaphysical identification of mathematics with physics simpliciter.

1. Every primitive of *the mathematics that physics uses*, from the natural numbers through Coxeter groups, has been derived above from the substrate's primitive content. Each derivation is paired with an observable discharge.

2. The substrate's primitive content is, by the framework's implication-graph construction, traceable back to a single ontological root: the substrate exists.

3. Within that scope, mathematics is what falls out of the substrate's primitive content when that content is described formally; the bootstrap-derived objects and the standard-mathematics objects are isomorphic by the bridge theorems (Sec. II A). We do not extend this claim to mathematics outside the scope.

4. Applying that slice of mathematics to physics is therefore not a contingent mapping between two distinct domains. The formal description and the substrate-physical content it describes are the same object up to bridge-theorem isomorphism, and the description's correctness on the physics is a consequence of that isomorphism rather than a separately discovered fit.

The "unreasonable effectiveness" of mathematics in the natural sciences is, on this reading, the consequence of a structural identity rather than a coincidence between two independent objects. The mathematics that appears in successful physics is the formal description of substrate-physical content; the physics that mathematics describes successfully is the substrate's actual behaviour.

The empirical content of any physical theory — a measurement of an oscillation frequency, a charge, a decay width — is a substrate process running, and the mathematical description of that process is the formal capture of the same substrate content. The puzzle dissolves because the assumption it required — mathematics on one side, physics on the other, with a contingent mapping in between — is not a feature of the framework. The "mapping" is the formal description, and the description's correctness on the physics is bound to follow because the description is, by construction, that content's formalisation.

This explains, without further explanation, why the parts of mathematics that are most useful to physics are the parts that formalise substrate-physical content directly. Linear superposition is useful because the substrate has a linear regime. Differential forms are useful because the substrate's coupling events are oriented. Lie groups are useful because the substrate has continuous symmetries. Root systems are useful because the substrate's mode catalogue is quantised. Each is the formal description of an observed substrate-physical phenomenon, and the formal description is bound to be effective on the phenomenon it formalises.

It also explains, structurally, the cases that surprised Wigner. Complex numbers are not a calculational convenience that surprised us by being irreducible to quantum mechanics. The substrate's wave-pattern amplitudes carry an oriented two-dimensional structure (magnitude and phase) under which composition acts by rotation: multiplication by $e^{i\theta}$ rotates a substrate amplitude by angle θ in the amplitude plane, and in particular multiplication by i is the quarter-turn (so that $i^2 = -1$ is the half-turn that takes an amplitude to its negative, and $i^4 = 1$ returns it to identity). The complex numbers are the smallest field that supports this oriented rotational composition algebraically; their appearance wherever the substrate's amplitude phase matters (interference, gauge phase, Bloch states, the Schrödinger equation) is structural rather than coincidental. Hilbert spaces are similarly not an abstraction that surprised us by being the right home for quantum states. The substrate's standing-wave amplitudes form a vector space under linear superposition (which the substrate already commits to in its free-propagation regime), and the choice of inner product reflects the substrate's amplitude pairing under which $|\psi|^2$-type integrals — conserved quantities, transition probabilities, mode overlaps — are the appropriate physical observables. The completion of this vector space under the corresponding norm is a Hilbert space; the L^2 choice of pairing is what makes the substrate's continuity equation $\partial_t|\psi|^2 + \nabla \cdot \mathbf{j} = 0$ consistent with the unitary linear dynamics of free propagation. We do not claim Hilbert-space completeness or the L^2 choice are *forced* from substrate primitives alone in this paper; we claim the framework's free-propagation, superposition, and conserved-amplitude commitments together select Hilbert-space structure as their natural formal capture, which is why Hilbert spaces turn out to be where quantum mechanics lives.

VIII. LIMITS OF THE PRESENT CONSTRUCTION

We state the limits plainly. The bootstrap chain above is structural and observationally anchored at every step, but several pieces are provisional.

- *Selection of the specific algebra.* The chain produces Lie algebras and root systems in general but does not, in itself, force $E_8 \times E_8$ specifically. The E_8 identification is empirical, anchored in the Zamolodchikov spectrum measured in $CoNb_2O_6$ [7, 8]; the doubling to $E_8 \times E_8$ has separate empirical motivation in heterotic-string anomaly cancellation and the visible-vs-dark sector split of the companion paper's [12] mass spectrum. Upgrading the algebra selection from empirical to structurally forced would require a substrate-physical projection onto the heterotic moduli space — the same open work flagged in the companion paper.

- *Discharge completeness.* Most of the substrate-mathematical hypotheses admit immediate discharge from the substrate's coupling-event definition or direct observation. Two — abstraction congruence and the strongest form of type instantiation — are provisionally discharged pending the substrate's standing-wave-mode catalogue, which is supplied elsewhere in the framework [11] but is not part of the present bootstrap. The provisional discharges are sufficient for everything in the present paper.

- *Construction-and-bridge equivalence.* Each substrate-derived object has both an existence-form definition (anchored in observation) and a constructive form (mirroring standard mathematics). The two presentations are connected by bridge theorems showing they describe the same object. The bridge theorems are stated and machine-verified throughout the framework's formal companion [11]; the present paper takes them as given and refers the reader to the formal record.

- *Higher mathematics.* The bootstrap above runs from the natural numbers through Coxeter groups. Higher structures — algebraic geometry, sheaves, derived categories, ∞-toposes — have not been bootstrapped here. The framework's expectation is that they admit substrate-physical readings of the same character (sheaves as substrate-mode patterns over base manifolds, derived categories as the substrate's homological structure, etc.), but the present paper does not work that out.

- *Quantum mechanics, general relativity, and the SM.* The application of the bootstrap to specific physics frameworks is summarised in the companion paper [12] and developed in detail in [11]. The Standard Model parameter spectrum, Newton's constant, and a unified neutrino–dark-matter tower are derived from a single anchor (the electron rest mass) in that work; the present paper provides the foundational mathematics on which those derivations rest.

IX. CONCLUSION

Wigner's question — why is mathematics so unreasonably effective in physics? — admits a structural answer once the assumption underlying it is dropped. The assumption was that mathematics and physics are two different things and that the success of mathematics in physics is a coincidence requiring explanation. We have argued that, in a substrate-physical framework with a single ontological root (the substrate exists), every primitive of *the mathematics physics uses* — the slice enumerated in Sec. II A — is derivable from the substrate's primitive content and discharged by an observable phenomenon. Mathematics is not a separate body of structure that happens to fit physics; it is the formal description of the substrate's primitive content. Its effectiveness in physics is structural: the mathematics that appears in physics is, by construction, the formalisation of the substrate-physical content that the physics measures.

The construction is observational at every step. Each substrate-mathematical object listed has an anchoring phenomenon. Each step is paired with a discharge obligation that has been either directly satisfied or is in progress. The framework's implication graph terminates, transitively, at the single root commitment that the substrate exists. We do not claim the construction is complete — the open items in Section VIII flag what remains — but the structure is in place, and the empirical anchor for the specific E_8 algebra has been observed in the laboratory.

If correct, the consequence is that for the slice of mathematics that physics actually uses, "unreasonable effectiveness" is a category mistake: that slice and the physics it describes are not two contingently related domains but two levels of formalisation of the same substrate-physical content. We do not extend the claim beyond that slice. Mathematics whose objects have no observed substrate-physical instantiation — large cardinals, certain choice-dependent constructions, exotic foundations, and so on (Sec. II A) — is consistent with the present framework but is not claimed to be the substrate's formalisation. The dissolution of Wigner's puzzle is therefore narrow and load-bearing rather than universal: the parts of mathematics that fit physics fit because they describe the substrate, and the parts that do not fit are unaffected by the present argument.

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The framework summarised here was developed across many threads of research collected in [11]; the bootstrap material draws on the foundational mathematics treatment in that work. The companion paper [12] carries out the quantitative-physics applications that the present paper grounds.

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